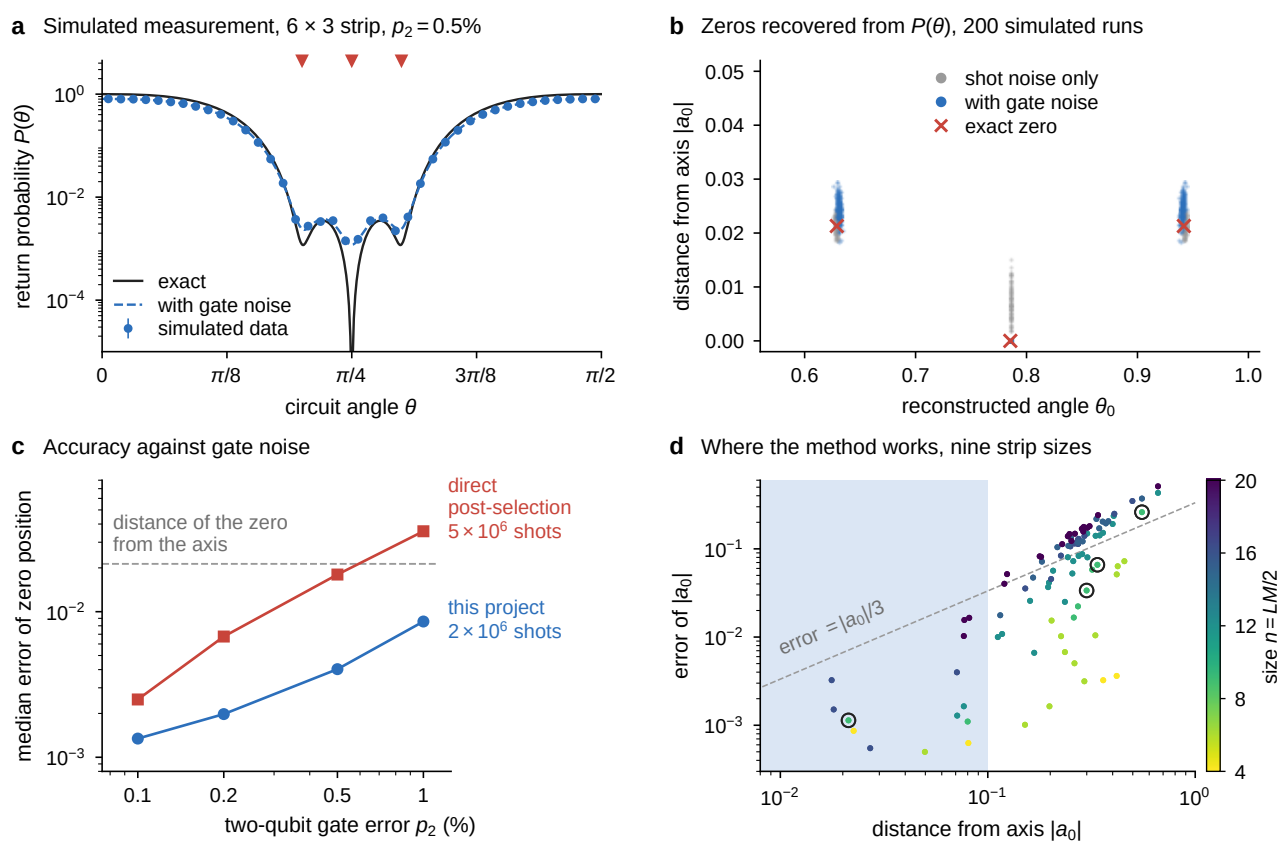
a
Simulated measurement, 6 × 3 strip, $p_2 = 0.5\%$
return probability $P(\theta)$
10^0
10^{-2}
10^{-4}
exact
with gate noise
simulated data
0
$\pi/8$
$\pi/4$
$3\pi/8$
$\pi/2$
circuit angle θ
b
Zeros recovered from $P(\theta)$, 200 simulated runs
shot noise only
with gate noise
exact zero
distance from axis $|a_0|$
0.05
0.04
0.03
0.02
0.01
0.00
0.6
0.7
0.8
0.9
1.0
reconstructed angle θ_0
c
Accuracy against gate noise
median error of zero position
distance of the zero
from the axis
direct
post-selection
5×10^6 shots
this project
2×10^6 shots
10^{-2}
10^{-3}
0.1
0.2
0.5
1
two-qubit gate error p_2 (%)
d
Where the method works, nine strip sizes
error of $|a_0|$
error $= |a_0|/3$
10^{-1}
10^{-2}
10^{-3}
10^{-2}
10^{-1}
10^0
distance from axis $|a_0|$
size $n = LM/2$
20
16
12
8
4